

Abigail Kapocius

South Bend, IN

akapociu@nd.edu

919-723-8555

EDUCATION

University of Notre Dame

Ph.D. in Computer Science and Engineering

South Bend, IN

August 2025 - Present

University of North Carolina at Charlotte

Bachelor of Science in Computer Science, Minor in English Literature

Charlotte, NC

Aug 2021 - May 2025

EXPERIENCE

Graduate Research Assistant — Weninger Lab

Physics-Inspired Temporal Graph Learning and Neural Interaction Modeling

South Bend, IN

January 2026 - Present

- **Physics-Inspired Temporal Graph Neural Networks** : Implemented and evaluated modular temporal graph learning architectures that combine physics-inspired and memory-based update rules and neural message aggregation for interaction prediction and future physical-state regression from complete and partially observed event streams.
- **Physical-System and Social Temporal Network Benchmarking** : Generated temporal interaction benchmarks from synthetic physical systems, molecular dynamics trajectories, and JODIE interaction datasets to analyze how graph structure, persistence, corruption, and model architecture influence edge prediction and node-level dynamics forecasting.
- **Experimental Pipeline Development** : Designed reproducible temporal graph learning pipelines spanning dataset construction, model configuration, hyperparameter tuning, training/evaluation automation, metric reporting, baseline comparison, and statistical analysis.
- **Hopfield-Inspired Graph Transformer Masking**: Developed a Hopfield-inspired evaluation framework for graph transformer masks, treating masks as constraints on attention-based associative retrieval and analyzing how topology, sparsity, receptive field size, and corruption affect candidate access, retrieval quality, and downstream node classification behavior.

Intern — Oak Ridge National Laboratory

Large-scale Graph Analysis and GPU Profiling/Optimization

Knoxville, TN

June 2024 - April 2025

- **Distributed Graph Analytics on Leadership Supercomputers**: Experimented with 70+ real-world and synthetic graph datasets, including graphs up to 4.2 billion edges, and analyzed structural properties such as degree distribution and density across biology, social, road, and citation networks.
- **Community Detection Experiments**: Evaluated HipMCL and HiPDPL-GPU for distributed community detection on Frontier and Summit, tuning runtime parameters and analyzing scalability, performance limits, and output quality using NMI, conductance, and modularity.
- **GPU Profiling and Optimization**: Profiled GPU-accelerated BFS and SSSP implementations in Gunrock using HPCToolkit, ROCm Profiler, NVIDIA Nsight Systems, and Nsight Compute to analyze kernel performance, memory usage, and GPU activity across AMD and NVIDIA systems.
- **Workshop Paper Contributions**: Contributed to an *IEEE International Conference on Big Data 2024* workshop paper by writing background, experimental setup, result visualizations, conclusions, and comparative performance analysis.
- **Leadership Supercomputing Architectures** : Gained hands-on experience with Frontier and Summit architectures, including AMD EPYC, IBM POWER9, AMD MI250X, and NVIDIA V100 hardware, using MPI, OpenMP, and HIP for scalable graph analytics workloads.

Undergraduate Research Assistant — High Performance Computing Systems Lab

Visualization and Profiling of I/O Access Patterns

Charlotte, NC

Jan 2023 - May 2023

- **I/O Access Pattern Analysis**: Analyzed Darshan logs from supercomputers Cori and Summit to identify I/O access patterns, bottlenecks, and contention affecting HPC application performance.
- **Independent Learning in HPC**: Developed foundational knowledge of HPC systems, including file systems, parallelization, Linux environments, and I/O operations.
- **Proficiency in HPC Technologies**: Applied HPC tools and interfaces including HDF5, MPIIO, POSIX, and Darshan for I/O analysis and profiling tasks.
- **Programming Skills**: Used Python and shell scripting to automate data processing and analyze large-scale HPC profiling datasets.

Undergraduate Research Assistant — Undergraduate Research Initiative

Charlotte, NC

Visualization and Profiling of I/O Access Patterns

May 2023 - Aug 2023

- **Enhanced Performance Analysis Tool:** Developed visualizations and graphs for integration into the Drishti performance analysis tool to improve clarity for application optimization.
- **I/O Profiling Techniques:** Created visualizations of HPC I/O profiling data to help users interpret performance behavior and optimization opportunities.
- **In-Depth Darshan Documentation Study:** Studied Darshan documentation to understand profiling capabilities and extract meaningful performance insights from I/O logs.

Peer Tutor — College of Computing and Informatics

Charlotte, NC

Logic and Algorithms

Aug 2022 - Nov 2024

- **Teaching:** Supported students with difficult concepts, assignments, and test preparation in introductory computer science coursework.
- **Logic and Algorithms/Introduction to Computer Science:** Led tutoring sessions for Logic and Algorithms and Introduction to Computer Science, helping new students build confidence in early CS topics.

PUBLICATIONS AND PRESENTATIONS

Naw Safrin Sattar, **Abigail Kapocius**, et al. "Exploring the Landscape of Distributed Graph Clustering on Leadership Supercomputers." *IEEE International Conference on Big Data 2024 Workshops*, Washington, D.C., December 2024.

Abigail Kapocius, Naw Safrin Sattar. "Evaluating Distributed Community Detection Methods for Large Graphs on Leadership Supercomputers." *DOE Cybersecurity and Technology Innovation Conference*, Dallas, TX, June 2024. Poster presentation.

Abigail Kapocius, Naw Safrin Sattar. "Evaluating Distributed Community Detection Methods for Large Graphs on Frontier Supercomputer." *IEEE International Conference on Big Data*, Washington, D.C., December 2024. Poster presentation.

Abigail Kapocius, Naw Safrin Sattar. "Evaluating Parallel GPU Implemented Shortest Path Algorithms for Large Graphs on Frontier Supercomputer" *Oak Ridge National Lab*, Knoxville, TN, April 2025. Poster presentation.

Abigail Kapocius "No, Not That Kind of Graph!" *Oak Ridge National Lab*, Knoxville, TN, April 2025. Ignite Talk presentation.

ADDITIONAL ACTIVITIES

Social Development Chair for the Graduate Society of Women Engineers: Elected for the 2026-2027 term to organize social and professional community-building events for graduate women in engineering for the University of Notre Dame chapter of GradSWE, fostering peer connection, interdisciplinary networking, and an inclusive support network across engineering departments.

Student Volunteer at SC24: Selected as a student volunteer for SC24, assisting with conference operations while gaining direct exposure to cutting-edge developments in HPC and building connections with leading professionals in the field.

HPC Immersion Program at SC23: Selected as one of twenty worldwide participants for the HPC Immersion Program at SC23, a program aimed at providing in-depth exposure to high-performance computing (HPC) for passionate undergraduate students, offering enhanced knowledge and opportunities.

HPC Basics Certification: Successfully completed a crash course, conducted by Oak Ridge National Laboratory and utilizing the Frontier supercomputer, focused on enhancing technical skills in an HPC environment with a particular emphasis on supporting students (completed in Nov 2023).

PROGRAMMING SKILLS

Languages: Python, C, Bash

Machine Learning and Data Science: PyTorch, PyTorch Geometric, scikit-learn, NumPy, pandas, Matplotlib, Jupyter, graph neural networks, temporal graph learning, representation learning, model evaluation, hyperparameter tuning

High-Performance Computing: CUDA, HIP, MPI, OpenMP, SLURM, parallel computing, GPU programming, distributed-memory computing

Performance Analysis and Storage Systems: HPCToolkit, Rocprof, Nsight Compute, Nsight Systems, Darshan, Drishti, Lustre, parallel file systems, high-performance storage systems